CLab 1: GNS3

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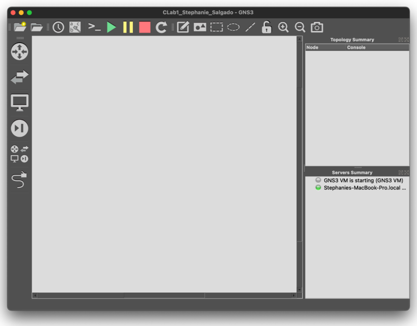
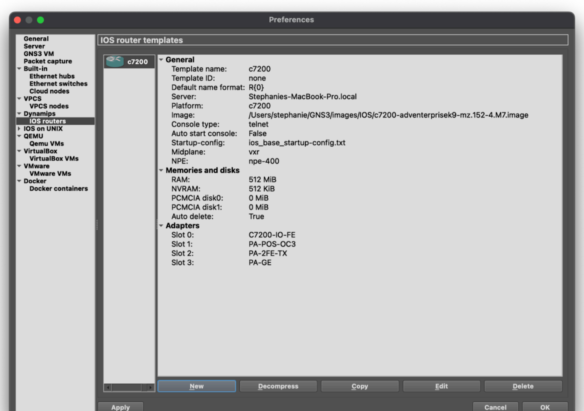
Installing GNS3



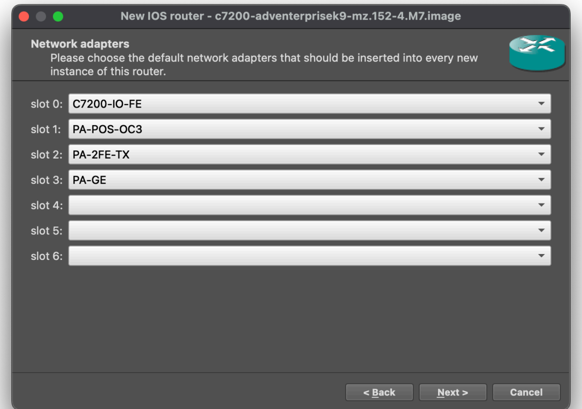
To install the software, you must create an account then download the appropriate version based on your OS. Then, I followed the installation steps.

Adding the Cisco C7200 router image to GNS3



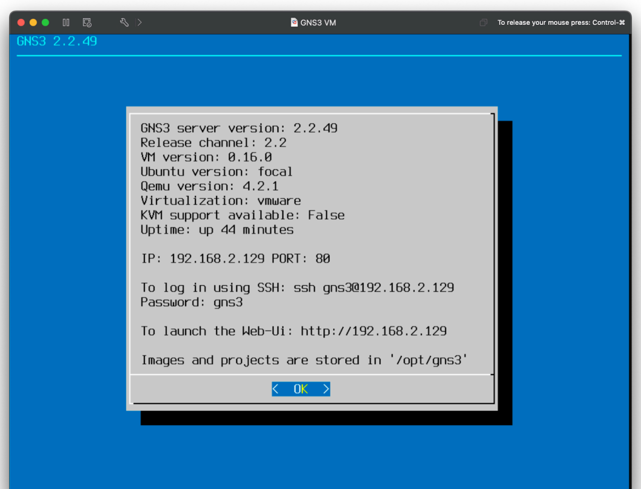
To add the router image, you navigate to “Preferences” > “Dynamips” > “IOS Routers” > “New”. Once on that screen, you can browse for your image. There are a few more steps to complete then, the image is added.

Configuring the router

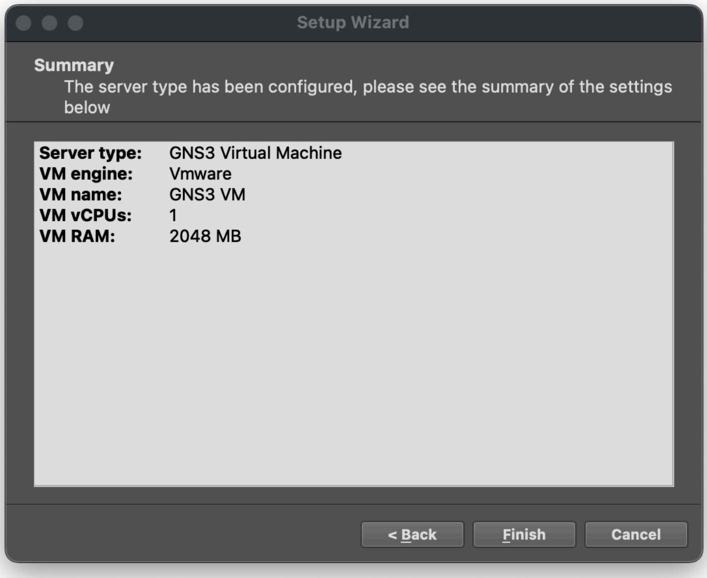


Once the router is added, it can be configured with different interfaces and slots using the "Edit" button.

Linking GNS3 VM and GNS3

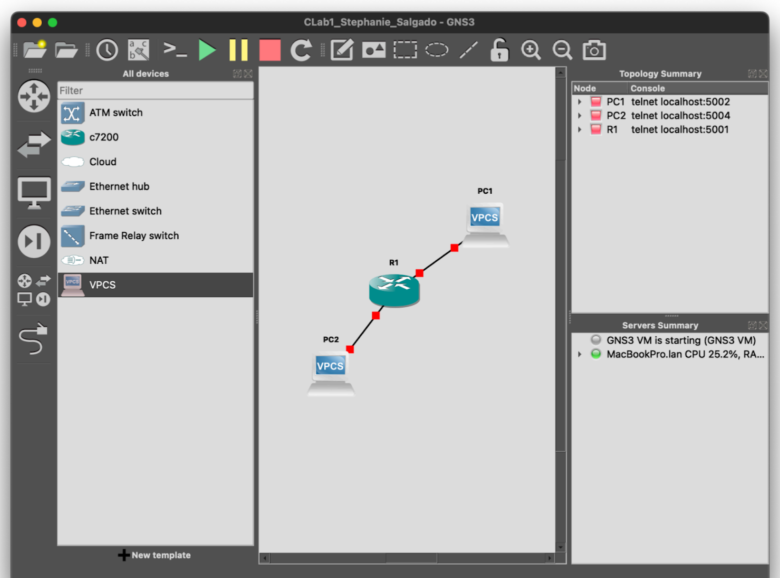


I used VMware Fusion. To add the GNS3 VM, I launched VMware and added a new virtual machine then navigated to the location of the GNS3 VM files I downloaded. I changed some settings and got it to work.

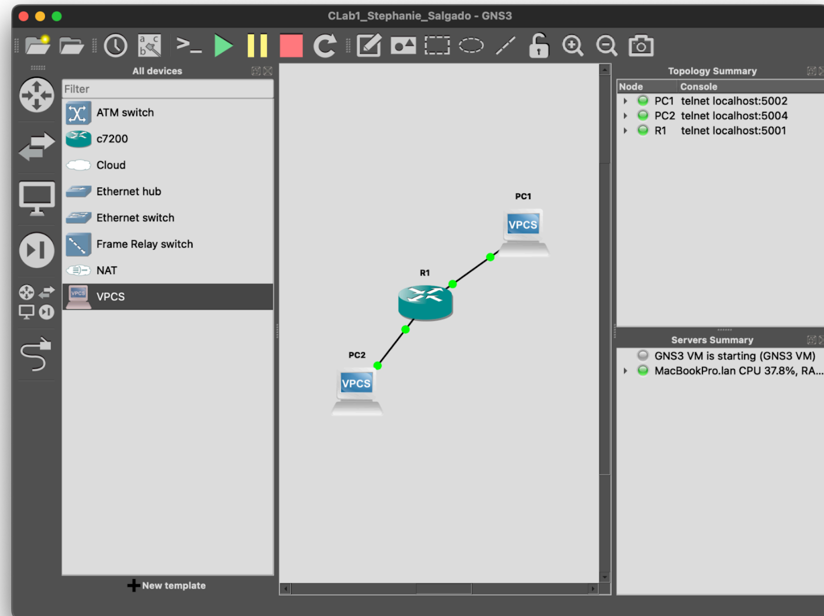


To add the VM, I simply launched GNS3 and opened the "Setup Wizard".

Creating a small topology



To do this, I added a router (R1) and 2 VPCS (PC1 and PC2). Then I added a link.



Then, I started all devices.

After that, I clicked on the console icon to console connect to all nodes.

R1 to PC1

```
R1#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
R1(config)#interface f0/0
R1(config-if)#ip address 10.1.1.1 255.255.255.0
R1(config-if)#no shutdown
R1(config-if)#end
R1#
*Aug 30 17:22:24.787: %SYS-5-CONFIG_I: Configured from console by console
R1#
*Aug 30 17:22:25.391: %LINK-3-UPDOWN: Interface FastEthernet0/0, changed state to up
*Aug 30 17:22:26.391: %LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0, changed state to up
R1#
```

R1 to PC2

```
R1#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
R1(config)#interface f1/0
R1(config-if)#ip address 10.1.2.1 255.255.255.0
R1(config-if)#no shutdown
R1(config-if)#end
R1#
*Aug 30 17:24:42.691: %SYS-5-CONFIG_I: Configured from console by console
R1#
```

Configuring the 2 hosts

```
PC1> dhcp
DDD
Can't find dhcp server

PC1> ip 10.1.1.30 255.255.255.0 10.1.1.3
Checking for duplicate address...
PC1 : 10.1.1.30 255.255.255.0 gateway 10.1.1.3

PC1> save
Saving startup configuration to startup.vpc
.  done

PC1>
```

```
PC2> dhcp
DDD
Can't find dhcp server

PC2> ip 10.1.1.40 255.255.255.0 10.1.1.3
Checking for duplicate address...
PC1 : 10.1.1.40 255.255.255.0 gateway 10.1.1.3

PC2> ping 10.1.1.30
host (10.1.1.30) not reachable

PC2>
```

Show ip route

```
R1#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
       + - replicated route, % - next hop override

Gateway of last resort is not set

      10.0.0.0/8 is variably subnetted, 4 subnets, 2 masks
C        10.1.1.0/24 is directly connected, FastEthernet0/0
L        10.1.1.1/32 is directly connected, FastEthernet0/0
C        10.1.2.0/24 is directly connected, FastEthernet1/0
L        10.1.2.1/32 is directly connected, FastEthernet1/0
R1#
```

Show ip

```
PC1> show ip

NAME        : PC1[1]
IP/MASK     : 10.1.1.30/24
GATEWAY     : 10.1.1.3
DNS         :
MAC         : 00:50:79:66:68:00
LPORT       : 10010
RHOST:PORT  : 127.0.0.1:10011
MTU:        : 1500

PC1>
```

```
PC2> show ip

NAME        : PC2[1]
IP/MASK     : 10.1.1.40/24
GATEWAY     : 10.1.1.3
DNS         :
MAC         : 00:50:79:66:68:01
LPORT       : 10008
RHOST:PORT  : 127.0.0.1:10009
MTU:        : 1500

PC2>
```

Show ip interface brief

```
R1#show ip interface brief
Interface                  IP-Address      OK? Method Status                Protocol
FastEthernet0/0            10.1.1.1        YES manual up                    up
FastEthernet1/0            10.1.2.1        YES manual up                    up
```

Repeat

```
R1#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
R1(config)#interface f0/0
R1(config-if)#ip address 192.168.1.2 255.255.255.0
R1(config-if)#no shutdown
R1(config-if)#end
R1#
*Aug 30 17:53:00.547: %SYS-5-CONFIG_I: Configured from console by console
R1#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
R1(config)#interface f1/0
R1(config-if)#ip address 192.168.100.1 255.255.255.0
R1(config-if)#no shutdown
R1(config-if)#end
R1#
*Aug 30 17:53:37.191: %SYS-5-CONFIG_I: Configured from console by console
R1#
```

```
R1#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
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C        10.1.1.0/24 is directly connected, FastEthernet0/0
L        10.1.1.1/32 is directly connected, FastEthernet0/0
C        10.1.2.0/24 is directly connected, FastEthernet1/0
L        10.1.2.1/32 is directly connected, FastEthernet1/0
R1#show ip interface brief
Interface                  IP-Address      OK? Method Status                Protocol
FastEthernet0/0            10.1.1.1        YES manual up                    up
FastEthernet1/0            10.1.2.1        YES manual up                    up
FastEthernet2/0            unassigned      YES unset  administratively down down
FastEthernet3/0            unassigned      YES unset  administratively down down
GigabitEthernet4/0         unassigned      YES unset  administratively down down
R1#
```

Result

```
PC2> show ip

NAME         : PC2[1]
IP/MASK      : 10.1.2.30/24
GATEWAY      : 10.1.2.1
DNS          :
MAC          : 00:50:79:66:68:01
LPORT        : 10008
RHOST:PORT   : 127.0.0.1:10009
MTU:         : 1500

PC2> ping 10.1.1.1
84 bytes from 10.1.1.1 icmp_seq=1 ttl=255 time=15.350 ms
84 bytes from 10.1.1.1 icmp_seq=2 ttl=255 time=15.439 ms
84 bytes from 10.1.1.1 icmp_seq=3 ttl=255 time=15.114 ms
84 bytes from 10.1.1.1 icmp_seq=4 ttl=255 time=15.346 ms
84 bytes from 10.1.1.1 icmp_seq=5 ttl=255 time=15.867 ms

PC2> ping 10.1.1.30
10.1.1.30 icmp_seq=1 timeout
10.1.1.30 icmp_seq=2 timeout
10.1.1.30 icmp_seq=3 timeout
10.1.1.30 icmp_seq=4 timeout

PC2> ping 10.1.2.1
84 bytes from 10.1.2.1 icmp_seq=1 ttl=255 time=15.397 ms
84 bytes from 10.1.2.1 icmp_seq=2 ttl=255 time=15.338 ms
84 bytes from 10.1.2.1 icmp_seq=3 ttl=255 time=15.594 ms
84 bytes from 10.1.2.1 icmp_seq=4 ttl=255 time=15.883 ms
```

I was able to get the PC to ping its gateway and the other machine on the other interface of the router.

Summary

In this lab, I familiarized myself with GNS3 and created a small topology. Then, I configured the router and 2 VPCS. I tried this lab on my M1 Mac using VMware Fusion and GNS3 for MacOS but encountered issues so I switched to my Windows desktop. I used VirtualBox in this case. It was very interesting to try both ways. Ultimately I found success when trying this lab on Windows.